Hongrui (Sam) Sheng

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EDUCATION

Fudan University (FDU) **Shanghai, China**
Bachelor of Science in Chemistry, GPA: 3.85/4.00 | Rank: 8/99 *09/2023 - Expected 06/2027*
· Key Coursework: Quantum Chemistry (Honor): A+; Organic Chemistry: A; Physical Chemistry: A

Stanford University **Stanford, CA, USA**
International Honors Program Participant, GPA: 4.11/4.30 *06/2025 - 08/2025*
· Key Coursework: Introduction to Probability Theory: A+; Programming Abstractions: A

PUBLICATIONS

1. P. Walther†, **H. Sheng**†, X. Liu†, B. Feng, R. Coyle, X. Yan, K. Smith, H. Kayal, S. C. Pal, and Z. Zheng*, "Predicting Scale-Up of Metal-Organic Framework Syntheses with Large Language Models," *arXiv preprint*, 2026. https://arxiv.org/abs/2604.20899
2. C. Lei, Y. Meng, **H. Sheng**, S. Wang*, X. Wang, and M. Gong*, "The Interplay of Ions and Surfaces during Deposition/Dissolution Equilibrium Enables Efficient Neutral-pH Biomass Electrooxidation," *submitted for publication*, 2026.

† Equal contribution; * Corresponding author.

RESEARCH EXPERIENCE

Predictive Synthesis of Metal-Organic Frameworks (MOFs) **St. Louis, MO, USA**
Visiting Researcher | Professor Zhiling (Zach) Zheng, Washington University in St. Louis *04/2026 - present*

- **ScaleMOF: Predicting Scale-up of MOF Syntheses**
  – Drafted the Supporting Information and designed core figures to visualize workflows and benchmark metrics.
  – Spearheaded the resubmission process, resolving a critical bug in negative sample splitting to guarantee objective and comprehensive model evaluation.
  – Re-benchmarked fine-tuned LLMs against text-embedding-augmented ML baselines and validated chemical diversity via t-SNE.
- **ActMOF: A 6-Million-Condition Benchmark for Active Learning in MOF Syntheses**
  – Benchmarked Active Learning algorithms across a 6-million-condition chemical space to optimize experimental efficiency in MOF syntheses.
  – Performed high-dimensional topological analysis and designed publication-quality visualizations.
- **BayesianMOF: Bayesian Modeling for MOF Adsorption**
  – Engineered an automated Python pipeline to integrate LLM-mined literature metadata with geometric descriptors.
  – Developed Bayesian Inference model to predict MOF adsorption behavior, replacing deterministic point-estimates with robust uncertainty quantification.

AI-Guided Bimetallic Electrocatalysts for Glycerol-to-Lactic Acid Conversion **Shanghai, China**
Independent Research Leader, Wangdao Research Program | Professor Ming Gong, FDU *05/2025 - present*

- Curated literature datasets of 52 entries and engineered β function-inspired biasing factors to encode complex chemical interactions for downstream model training.
- Benchmarked ML models (GBR, RF) coupled with heuristic optimization (GA, PSO) to screen 1 million metal alloy combinations, identifying candidates with a predicted lactic acid selectivity of $\sim$70%.
- Validated computational predictions through wet-lab synthesis, successfully achieving a peak experimental lactic acid selectivity of 56.55% with electrocatalyst $Pt_{66}Cu_{34}$.
- Optimized reaction conditions further by evaluating electrolyte effects (NaOH) and pulse electrolysis to increase lactic acid selectivity by $\sim$7%.

Biomass Valorization via Electrocatalysis **Shanghai, China**
Research Assistant | Professor Ming Gong, FDU *03/2024 - 04/2025*

- Synthesized catalysts and conducted electrochemical testing for the targeted electrooxidation of 5-hydroxymethylfurfural (HMF) to 2,5-furandicarboxylic acid (FDCA) under neutral conditions.
- Investigated the catalytic role of Co^{2+} and quantified product yields using UV-Vis spectroscopy and HPLC, applying foundational laboratory techniques to troubleshoot downstream product isolation and purification challenges.

SELECTED PROJECTS

AI-Driven Molecular Modeling and Simulation | **Codes on GitHub** — **Shanghai, China**
AI in Computational Chemistry and Simulation, Course Project — *03/2026 - 06/2026*

- Applied equivariant neural networks (PaiNN) and diffusion models (GeoDiff) using PyTorch and e3nn to predict atomic forces and generate 3D molecular conformations.
- Programmed custom Nosé-Hoover NVT integrators and Stochastic Surface Walking (SSW) algorithms via Python and LASP to explore complex potential energy surfaces and thermodynamic transitions.
- Fine-tuned Chemformer for single-step retrosynthesis, deploying an interactive web application, and applied unsupervised clustering (PCA, t-SNE) on QM9 datasets to map molecular properties.

Automated Computational Workflow for S_N2 Mechanisms | **Codes on GitHub** — **Shanghai, China**
Computational Chemistry, Course Project — *11/2025 - 12/2025*

- Automated Python-Gaussian workflows for 7 different amine methylation, engineering self-correcting protocols to resolve optimization failures in sterically hindered outliers.
- Applied SISSO-inspired regression ($R^2 > 0.95$) linking extracted electronic (Q_N) and topological (κ_3) descriptors to activation energies.

PROFESSIONAL ACTIVITIES & AFFILIATIONS

Peer Reviewer (Co-reviewer): *Nature Communications*, 2026
Member: American Chemical Society (ACS), American Institute of Chemical Engineers (AIChE), 2026

LEADERSHIP

Fudan Integrative Medicine Association & Traditional Chinese Medicine Club — **Shanghai, China**
Founding/Executive President — *03/2024 - 09/2025*

- Established the first student organization promoting interdisciplinary medicine at Fudan.
- Designed automated outreach workflows using Python, expanding membership to 100+ students.

HONORS & AWARDS

Outstanding Student, Fudan University, 2025
Second Prize, Syensqo Scholarship, Syensqo (formerly Solvay), 2025
Second Prize, Scholarship for Outstanding Students, Fudan University, 2024, 2025

SKILLS & ADDITIONAL INFORMATION

Programming & Tools	Python, LaTeX, C++; Computational Chemistry (Psi4, Gaussian, ASE, MACE)
Languages	Chinese (Native), English (TOEFL iBT 107/120), Japanese (JLPT N1)